# Modelling a 3D Printer

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# 0 Introduction

3D printers have the basic task of extruding filament at a given x, y, z coordinate at time t. The mechanism that achieves that movement determines the machine's costs, print quality, and robustness. Out of many different options, the Core XY mechanism is a popular choice for modern printers. Unlike Cartesian designs, where one motor drives one axis, here each motor simultaneously changes the x and y position of the tool head (see Figure 1). This results in a more complex kinematic model, but allows a sturdier casing and faster printing speeds. Our task is to model and simulate the system.

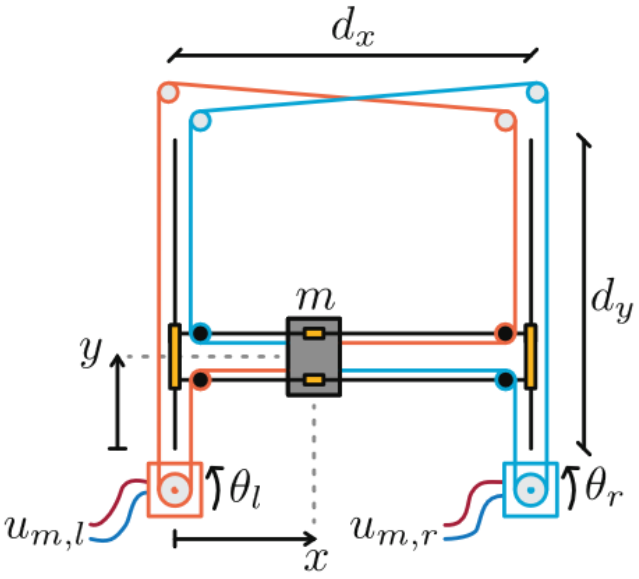



Figure 1: Schematic of the Core XY mechanism.

## 0.1 Symbol Table

Table 1: Symbols and parameter values.

| Description | Symbol | Value |
|---|---|---|
| Motor angle left | $\theta_L$ | – |
| Motor voltage left | $u_{m,L}$ | – |
| Motor current left | $i_L$ | – |
| Motor radius | $r_m$ | 11 mm |
| Motor inductance | $L_m$ | 10 mH |
| Motor resistance | $R_m$ | 90 Ω |
| Motor inertia | $J_m$ | $8 \times 10^{-6}$ kg m$^2$ |
| Motor constant | $K_m$ | 0.07 N m/A |
| Tool head position | $x$, $y$ | – |
| Tool head mass | $m$ | 60 g |
| Nozzle voltage | $u_n$ | – |
| Nozzle inductor current | $i_{L,n}$ | – |
| Nozzle capacitance | $C_n$ | – |
| Nozzle inductance | $L_n$ | – |
| Capacitor voltage | $u_c$ | – |
| Nozzle thermal capacity | $C_{t,n}$ | – |
| Nozzle thermal resistance | $R_{t,n}$ | 9 K/W |
| Nozzle resistance | $R_n$ | 8 Ω |
| Resistor thermal capacity | $C_r$ | – |
| Resistor temperature | $T_r$ | – |
| Thermal resistance to environment | $R_{\text{env}}$ | – |
| Reference resistance | $R_0$ | – |
| Reference temperature | $T_0$ | – |
| Temperature coefficient of resistance | $\alpha$ | – |
| Nozzle temperature | $T_n$ | 240 °C |
| Environmental temperature | $T_e$ | 26.85 °C |
| Bearing friction rolling | $\mu_r$ | $2.7 \times 10^{-8}$ N m s/rad |
| Bearing friction linear $x$-direction | $\mu_{l,x}$ | 0.002 N s/m |
| Bearing friction linear $y$-direction | $\mu_{l,y}$ | 0.001 N s/m |
| Bearing radius | $r_b$ | 3 mm |
| Print bed temperature | $T_b$ | 105 °C |
| Print area dimension $x$ | $d_x$ | 29 cm |
| Print area dimension $y$ | $d_y$ | 29 cm |

# 1 Physical Modelling

## 1.1 Kinematics

The first task is to derive the direct kinematics of the Core XY mechanism. This is slightly complicated by the fact that rotating one motor will move the tool head in both $x$ and $y$ directions. By removing the right motor, we can see that by moving the tool head in the positive $x$ direction **or** in the positive $y$ direction, the left motor will rotate in the positive direction. We can draw similar conclusions for the right motor, for which we get the following equations:

$$\begin{aligned} \Delta L &= \Delta x + \Delta y \\ \Delta R &= \Delta x - \Delta y \\ \Delta x &= \frac{\Delta L + \Delta R}{2} \\ \Delta y &= \frac{\Delta L - \Delta R}{2} \end{aligned} \tag{1.1}$$

With $\Delta L$ and $\Delta R$ being the length of belt moved by the motors, and $\Delta x$ and $\Delta y$ being the change in position of the tool head.

$$\begin{aligned} \Delta L &= r_m \, \theta_L \\ \Delta R &= r_m \, \theta_R \end{aligned} \tag{1.2}$$

If we assume that in the beginning the motors are at $\theta_L = \theta_R = 0$ and the tool head is at $(0, 0)$, we get the following equations:

$$\begin{aligned} x(\theta_L, \theta_R) &= \frac{r_m \theta_L + r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L + \theta_R\right) \\ y(\theta_L, \theta_R) &= \frac{r_m \theta_L - r_m \theta_R}{2} = \frac{r_m}{2} \left(\theta_L - \theta_R\right) \end{aligned} \tag{1.3}$$

$$\begin{aligned} \dot{x}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L + \dot{\theta}_R\right) \\ \dot{y}(\dot{\theta}_L, \dot{\theta}_R) &= \frac{r_m}{2} \left(\dot{\theta}_L - \dot{\theta}_R\right) \end{aligned} \tag{1.4}$$

## 1.2 Dynamics

In the next part we model the dynamics of the 3D printer system by applying the Euler–Lagrange equation. Additionally, we will split the problem into a mechanical and an electrical one, with the angular velocity of the motor shafts connecting them.

Because of that, we choose generalised coordinates:

$$q = \begin{bmatrix} x \\ y \\ q_L \\ q_R \end{bmatrix} \qquad \dot{q} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix} \tag{1.5}$$

We can also then split them among our two problems:

$$\dot{q}_m = \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix} \text{ (mechanical part)}, \qquad \dot{q}_e = \begin{bmatrix} i_L \\ i_R \end{bmatrix} \text{ (electrical part)} \tag{1.6}$$

### 1.2.1 Mechanical Part

#### Kinetic Energy

Firstly, we derive the kinetic energy terms. In the *mechanical* part, the only mass is the tool head. With that we have:

$$K_{\text{mech}} = \tfrac{1}{2} m(\dot{x}^2 + \dot{y}^2) \tag{1.7}$$

As the whole system is planar, we can see that there are no potential energy terms as it will be constant.

#### Dissipative Energy

Secondly, we find the dissipative energy terms. Each motor has 4 rotational bearings, and 4 linear bearings for the whole system, two along each axis. We define $r_b$ as the bearing radius, later it will be kept the same as the motor radius for simplicity. From (1.3), we can get transformations (1.8) with trivial time derivatives.

This needs to be corrected, not all bearings roll at all times, they depend on what motor rotates

$$\theta_{b,L} = \frac{x+y}{r_b}, \qquad \theta_{b,R} = \frac{x-y}{r_b} \tag{1.8}$$

Then, for each side we get rotational dissipative energy:

$$D_L = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,L}^2 \, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 + 2\dot{x}\dot{y} + \dot{y}^2\right]$$
$$D_R = 4 \cdot \tfrac{1}{2} \cdot \dot{\theta}_{b,R}^2 \, \mu_r = 2\mu_r \frac{1}{r_b^2} \left[\dot{x}^2 - 2\dot{x}\dot{y} + \dot{y}^2\right]$$

Then simply combining them:

$$\boxed{D_{\text{rot}} = 4 \, \frac{\mu_r}{r_b^2} \left[\dot{x}^2 + \dot{y}^2\right]}$$

Similarly, we get the linear dissipative energy:

$$D_{\text{lin}} = 2 \cdot \tfrac{1}{2} \mu_{l,x} \, \dot{x}^2 + 2 \cdot \tfrac{1}{2} \mu_{l,y} \, \dot{y}^2 = \mu_{l,x} \, \dot{x}^2 + \mu_{l,y} \, \dot{y}^2$$

Then, combining the terms we get:

$$D_{\text{mech}} = \left[4 \, \frac{\mu_r}{r_b^2} + \mu_{l,x}\right] \dot{x}^2 + \left[4 \, \frac{\mu_r}{r_b^2} + \mu_{l,y}\right] \dot{y}^2 \tag{1.9}$$

### 1.2.2 Electrical Part

Now we model the energies of the electrical part of the system. Keep in mind that both motors are the same. Below we can see the equivalent circuit of a DC motor:

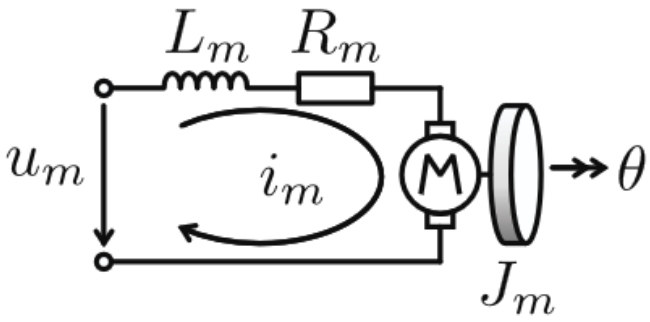



Figure 2: Equivalent circuit of a DC motor.

#### Kinetic Energy

The inductor and the inertial element will contribute to the kinetic energy of this part of the system. Similarly to (1.8), using the motor shaft radius:

$$K_L = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \dot{\theta}_L^2 = \tfrac{1}{2} i_L^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 + 2\dot{x}\dot{y}\right)$$
$$K_R = \tfrac{1}{2} i_R^2 L_m + \tfrac{1}{2} J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2 - 2\dot{x}\dot{y}\right)$$

Combining them we get:

$$K_{\text{el}} = \tfrac{1}{2} L_m \left[i_R^2 + i_L^2\right] + J_m \frac{1}{r_m^2} \left(\dot{x}^2 + \dot{y}^2\right) \tag{1.10}$$

As there are no capacitors in the system, there is no potential energy change as in the mechanical part.

#### Dissipative Energy

The only dissipative term in the electrical part of the system is the resistor:

$$\begin{gathered} D_{\text{el},L} = \tfrac{1}{2} i_L^2 R_m, \qquad D_{\text{el},R} = \tfrac{1}{2} i_R^2 R_m \\ D_{\text{el}} = \tfrac{1}{2} R_m \left(i_L^2 + i_R^2\right) \end{gathered} \tag{1.11}$$

#### Matrix Equations

Finally, we realise that the system is linear, and write the matrix expressions:

$$K = \tfrac{1}{2}\, \dot{q}^\top \mathbf{M}\, \dot{q}, \qquad D = \tfrac{1}{2}\, \dot{q}^\top \mathbf{N}\, \dot{q} \tag{1.12}$$

With corresponding matrices:

$$\mathbf{M} = \begin{bmatrix} m + J_m \frac{2}{r_m^2} & 0 & 0 & 0 \\ 0 & m + J_m \frac{2}{r_m^2} & 0 & 0 \\ 0 & 0 & L_m & 0 \\ 0 & 0 & 0 & L_m \end{bmatrix}$$
$$\mathbf{N} = \begin{bmatrix} 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x} & 0 & 0 & 0 \\ 0 & 8\frac{\mu_r}{r_b^2} + 2\mu_{l,y} & 0 & 0 \\ 0 & 0 & R_m & 0 \\ 0 & 0 & 0 & R_m \end{bmatrix} \tag{1.13}$$

### 1.2.3 Generalised Force

#### Mechanical Coordinates

The non-conservative forces in the mechanical part are given by the motor torques $\tau_L$ and $\tau_R$. The virtual work they perform is:

$$\Delta W = \tau_L \, \Delta\theta_L + \tau_R \, \Delta\theta_R$$

With the familiar transformation, analogous to (1.8) but using the motor radius $r_m$, we get:

$$\Delta W = \frac{\tau_L + \tau_R}{r_m} \, \Delta x + \frac{\tau_L - \tau_R}{r_m} \, \Delta y$$

Applying the motor law $\tau = K_m i$, the generalised forces for $x$ and $y$ are:

$$Q_x = \frac{K_m}{r_m} \left( i_L + i_R \right), \qquad Q_y = \frac{K_m}{r_m} \left( i_L - i_R \right) \tag{1.14}$$

#### Electrical Coordinates

The non-conservative forces in the electrical part are the input voltages and the back-EMF of the motors:

$$\Delta W = \left( u_{m,L} - K_m \dot{\theta}_L \right) \Delta q_L + \left( u_{m,R} - K_m \dot{\theta}_R \right) \Delta q_R$$

Using the same transformation, we get the generalised forces for $q_L$ and $q_R$:

$$Q_{q_L} = u_{m,L} - \frac{K_m}{r_m} \left( \dot{x} + \dot{y} \right), \qquad Q_{q_R} = u_{m,R} - \frac{K_m}{r_m} \left( \dot{x} - \dot{y} \right) \tag{1.15}$$

#### Lagrange Equations

Using the fact that this is a linear system and there are no potential energies we get:

$$\mathcal{L} = K - V = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q} - V, \qquad V = 0 \quad \Rightarrow \quad \mathcal{L} = \tfrac{1}{2} \, \dot{q}^\top \mathbf{M} \, \dot{q}$$

$$\frac{d}{dt} \frac{\partial \mathcal{L}}{\partial \dot{q}} = \frac{d}{dt} \left( \mathbf{M} \dot{q} \right) = \mathbf{M} \ddot{q}$$
$$\frac{\partial \mathcal{L}}{\partial q} = 0$$
$$\frac{\partial D}{\partial \dot{q}} = \mathbf{N} \dot{q}$$

And finally solving the Euler–Lagrange equation:

$$\frac{d}{dt}\frac{\partial \mathcal{L}}{\partial \dot{q}} - \frac{\partial \mathcal{L}}{\partial q} = -\frac{\partial D}{\partial \dot{q}} + \mathbf{Q}, \qquad \mathbf{Q} = \begin{bmatrix} Q_x \\ Q_y \\ Q_{q_L} \\ Q_{q_R} \end{bmatrix}$$

$$\mathbf{M}\ddot{q} + \mathbf{N}\dot{q} = \mathbf{Q} \tag{1.16}$$

Expanding back out of matrix form we get:

$$\begin{aligned}
\left(m + J_m \frac{2}{r_m^2}\right)\ddot{x} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}\right)\dot{x} - \frac{K_m}{r_m} i_L - \frac{K_m}{r_m} i_R &= 0 \\
\left(m + J_m \frac{2}{r_m^2}\right)\ddot{y} + \left(8\frac{\mu_r}{r_b^2} + 2\mu_{l,y}\right)\dot{y} - \frac{K_m}{r_m} i_L + \frac{K_m}{r_m} i_R &= 0 \\
L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\dot{x} + \frac{K_m}{r_m}\dot{y} - u_{m,L} &= 0 \\
L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\dot{x} - \frac{K_m}{r_m}\dot{y} - u_{m,R} &= 0
\end{aligned}$$

Or with substitutions, we get the final form of the four differential equations:

$$m_{\text{eq}} = m + \frac{2J_m}{r_m^2}, \qquad b_x = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,x}, \qquad b_y = 8\frac{\mu_r}{r_b^2} + 2\mu_{l,y}$$

$$\boxed{\begin{aligned}
\ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L + i_R\right) \\
\ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m}\left(i_L - i_R\right) \\
\frac{di_L}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} + \dot{y}\right) - \frac{R_m}{L_m} i_L + \frac{u_{m,L}}{L_m} \\
\frac{di_R}{dt} &= -\frac{K_m}{r_m L_m}\left(\dot{x} - \dot{y}\right) - \frac{R_m}{L_m} i_R + \frac{u_{m,R}}{L_m}
\end{aligned}} \tag{1.17}$$

Or in **state-space** form:

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \dot{x} \\ \dot{y} \\ i_L \\ i_R \end{bmatrix}, \qquad \dot{\mathbf{x}} = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \ddot{x} \\ \ddot{y} \\ \dfrac{di_L}{dt} \\ \dfrac{di_R}{dt} \end{bmatrix}, \qquad \mathbf{u} = \begin{bmatrix} u_{m,L} \\ u_{m,R} \end{bmatrix}$$

$$\dot{\mathbf{x}} = \mathbf{A}\,\mathbf{x} + \mathbf{B}\,\mathbf{u} \tag{1.18}$$

$$\mathbf{A} = \begin{bmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & -\frac{b_x}{m_{\text{eq}}} & 0 & \frac{K_m}{m_{\text{eq}} r_m} & \frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & -\frac{b_y}{m_{\text{eq}}} & \frac{K_m}{m_{\text{eq}} r_m} & -\frac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & -\frac{K_m}{r_m L_m} & -\frac{K_m}{r_m L_m} & -\frac{R_m}{L_m} & 0 \\ 0 & 0 & -\frac{K_m}{r_m L_m} & \frac{K_m}{r_m L_m} & 0 & -\frac{R_m}{L_m} \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ \frac{1}{L_m} & 0 \\ 0 & \frac{1}{L_m} \end{bmatrix}$$

### 1.2.4 Decoupling the $x$ and $y$ Dynamics

To decouple the system, we introduce a set of useful substitutions:

$$\boxed{\begin{aligned} i_x &= i_L + i_R, & u_{m,x} &= u_{m,L} + u_{m,R} \\ i_y &= i_L - i_R, & u_{m,y} &= u_{m,L} - u_{m,R} \\ i_L &= \frac{i_x + i_y}{2}, & u_{m,L} &= \frac{u_{m,x} + u_{m,y}}{2} \\ i_R &= \frac{i_x - i_y}{2}, & u_{m,R} &= \frac{u_{m,x} - u_{m,y}}{2} \end{aligned}} \tag{1.19}$$

By substituting our new decoupled currents we get:

$$\begin{aligned} \frac{di_L}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} + \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} - \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x - \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} + u_{m,y}\right) \\ \frac{di_R}{dt} &= \tfrac{1}{2}\frac{di_x}{dt} - \tfrac{1}{2}\frac{di_y}{dt} = -\frac{K_m}{r_m L_m}\dot{x} + \frac{K_m}{r_m L_m}\dot{y} - \frac{R_m}{2L_m} i_x + \frac{R_m}{2L_m} i_y + \frac{1}{2L_m}\left(u_{m,x} - u_{m,y}\right) \end{aligned}$$

Then by adding and subtracting the two equations to cancel terms we get:

$$\begin{aligned} \frac{di_x}{dt} &= -\frac{2K_m}{r_m L_m}\dot{x} - \frac{R_m}{L_m} i_x + \frac{1}{L_m} u_{m,x} \\ \frac{di_y}{dt} &= -\frac{2K_m}{r_m L_m}\dot{y} - \frac{R_m}{L_m} i_y + \frac{1}{L_m} u_{m,y} \end{aligned} \tag{1.20}$$

And by substituting (1.19) into the first two equations of (1.17):

$$\begin{aligned} \ddot{x} &= -\frac{b_x}{m_{\text{eq}}}\dot{x} + \frac{K_m}{m_{\text{eq}} r_m} i_x \\ \ddot{y} &= -\frac{b_y}{m_{\text{eq}}}\dot{y} + \frac{K_m}{m_{\text{eq}} r_m} i_y \end{aligned} \tag{1.21}$$

And finally:

$$\tilde{\mathbf{x}} = \begin{bmatrix} x \\ \dot{x} \\ i_x \\ y \\ \dot{y} \\ i_y \end{bmatrix}, \qquad \dot{\tilde{\mathbf{x}}} = \begin{bmatrix} \dot{x} \\ \ddot{x} \\ \dfrac{di_x}{dt} \\ \dot{y} \\ \ddot{y} \\ \dfrac{di_y}{dt} \end{bmatrix}, \qquad \tilde{\mathbf{u}} = \begin{bmatrix} u_{m,x} \\ u_{m,y} \end{bmatrix}$$

$$\dot{\tilde{\mathbf{x}}} = \tilde{\mathbf{A}}\,\tilde{\mathbf{x}} + \tilde{\mathbf{B}}\,\tilde{\mathbf{u}} \tag{1.22}$$

$$\tilde{\mathbf{A}} = \left[\begin{array}{ccc|ccc} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & -\dfrac{b_x}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} & 0 & 0 & 0 \\ 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & -\dfrac{b_y}{m_{\text{eq}}} & \dfrac{K_m}{m_{\text{eq}} r_m} \\ 0 & 0 & 0 & 0 & -\dfrac{2K_m}{r_m L_m} & -\dfrac{R_m}{L_m} \end{array}\right], \qquad \tilde{\mathbf{B}} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ \dfrac{1}{L_m} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & \dfrac{1}{L_m} \end{bmatrix}$$

## 1.3 Drawing Shapes

Rearranging the decoupled equations of motion (1.21), we can express the currents needed to follow a prescribed motion of the tool head:

$$\boxed{\begin{aligned} i_x &= \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{x} + \frac{b_x r_m}{K_m}\,\dot{x} \\ i_y &= \frac{m_{\text{eq}} r_m}{K_m}\,\ddot{y} + \frac{b_y r_m}{K_m}\,\dot{y} \end{aligned}} \tag{1.23}$$

The motor equations for each motor are:

$$\begin{aligned} u_{m,L} &= L_m \frac{di_L}{dt} + R_m i_L + K_m \dot{\theta}_L = L_m \frac{di_L}{dt} + R_m i_L + \frac{K_m}{r_m}\,(\dot{x} + \dot{y}) \\ u_{m,R} &= L_m \frac{di_R}{dt} + R_m i_R + \frac{K_m}{r_m}\,(\dot{x} - \dot{y}) \end{aligned}$$

Adding them, and using (1.19):

$$u_{m,x} = u_{m,L} + u_{m,R} = L_m \left[\frac{di_L}{dt} + \frac{di_R}{dt}\right] + R_m\,[i_L + i_R] + \frac{2K_m}{r_m}\,\dot{x}$$

and similarly by subtracting them, we get the decoupled input voltages, consistent with (1.20):

$$\boxed{\begin{aligned} u_{m,x} &= L_m \frac{di_x}{dt} + R_m i_x + \frac{2K_m}{r_m}\dot{x} \\ u_{m,y} &= L_m \frac{di_y}{dt} + R_m i_y + \frac{2K_m}{r_m}\dot{y} \end{aligned}} \tag{1.24}$$

The original motor voltages then follow from (1.19):

$$u_{m,L} = \tfrac{1}{2}\left(u_{m,x} + u_{m,y}\right), \qquad u_{m,R} = \tfrac{1}{2}\left(u_{m,x} - u_{m,y}\right) \tag{1.25}$$

Since (1.24) contains $\frac{di_x}{dt}$, and (1.23) contains $\ddot{x}$, the voltages depend on the third derivative of the position. Therefore, for any $C^3$ functions $x(t)$ and $y(t)$ describing the position of the tool head, we can derive an open-loop analytical solution for the input voltages, even if those voltages are not realistic.

**Ellipse**

An ellipse with centre $(h, k)$ and semi-axes $a$ and $b$ has the parametric form:

$$x(t) = h + a\cos t, \qquad y(t) = k + b\sin t \tag{1.26}$$

We also need the derivatives up to the third order:

$$\begin{aligned} \dot{x} &= -a\sin t, \quad \ddot{x} = -a\cos t, \quad \dddot{x} = a\sin t \\ \dot{y} &= b\cos t, \quad \ddot{y} = -b\sin t, \quad \dddot{y} = -b\cos t \end{aligned}$$

Substituting into (1.23):

$$\begin{aligned} i_x(t) &= -\frac{m_{\text{eq}} r_m}{K_m} a\cos t - \frac{b_x r_m}{K_m} a\sin t = -\frac{r_m}{K_m} a\left[m_{\text{eq}}\cos t + b_x \sin t\right] \\ i_y(t) &= \frac{r_m}{K_m} b\left[b_y \cos t - m_{\text{eq}}\sin t\right] \end{aligned}$$

And into (1.24):

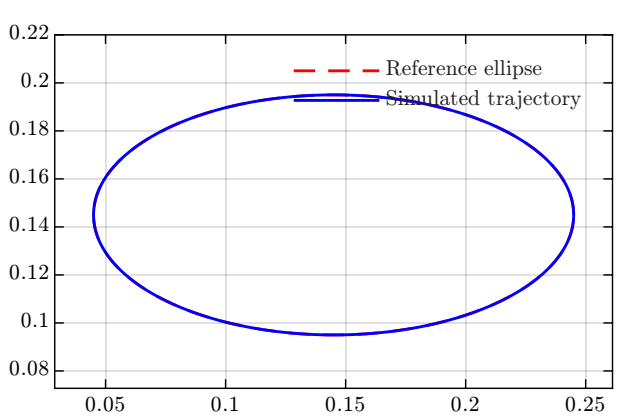
$$\begin{aligned} u_{m,x}(t) &= L_m \frac{r_m}{K_m} a\left[m_{\text{eq}}\sin t - b_x\cos t\right] - R_m \frac{r_m}{K_m} a\left[m_{\text{eq}}\cos t + b_x \sin t\right] - \frac{2K_m}{r_m} a\sin t \\ u_{m,y}(t) &= -L_m \frac{r_m}{K_m} b\left[m_{\text{eq}}\cos t + b_y \sin t\right] + R_m \frac{r_m}{K_m} b\left[b_y\cos t - m_{\text{eq}}\sin t\right] + \frac{2K_m}{r_m} b\cos t \end{aligned}$$

With

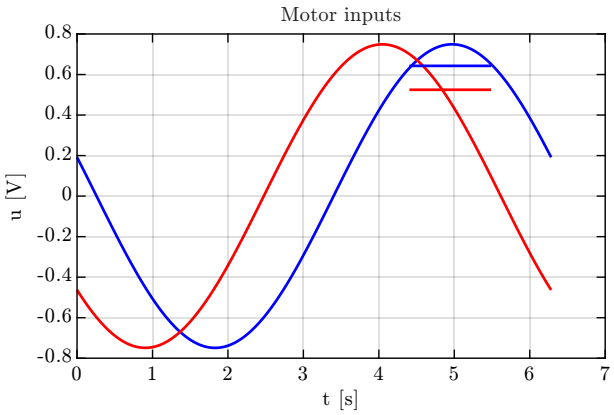
$$u_{m,L}(t) = \tfrac{1}{2}\left(u_{m,x}(t) + u_{m,y}(t)\right), \qquad u_{m,R}(t) = \tfrac{1}{2}\left(u_{m,x}(t) - u_{m,y}(t)\right)$$

we obtain the analytical open-loop solution for drawing the ellipse.

Simulating the given system with parameters defined in Table 1, we get near-perfect alignment with the reference ellipse.




(a) Reference and simulated trajectory.




(b) Open-loop motor voltages.

Figure 3: Open-loop drawing of an ellipse with nominal model parameters.

As this is open-loop control, this good result relies on a good parameter estimation. If we change the parameters of the model by ±5% we can see that an error immediately arises.

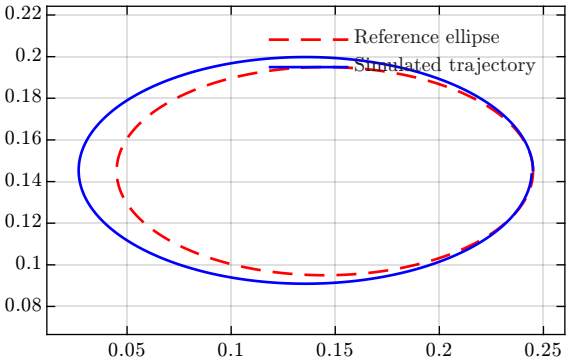
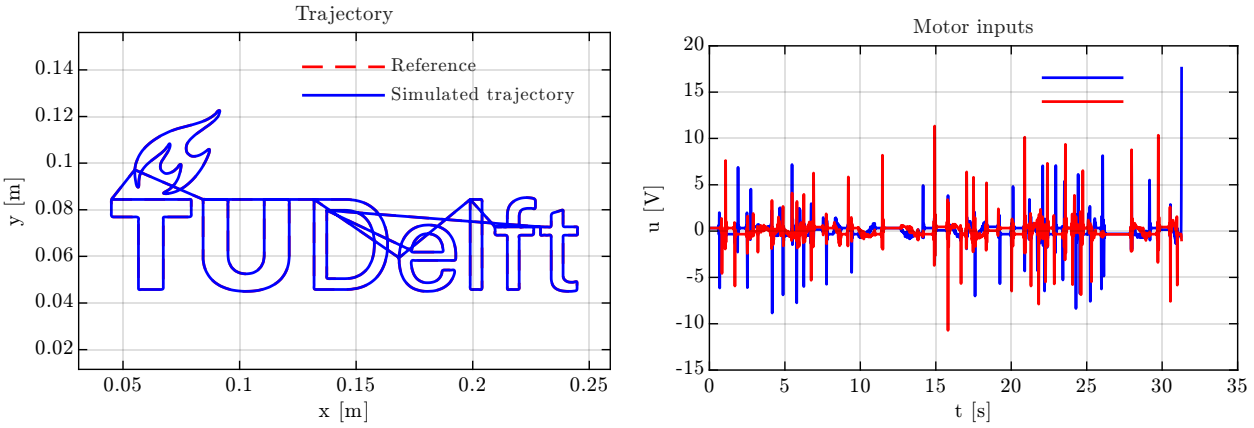



Figure 4: Open-loop drawing of an ellipse with model parameters perturbed by ±5%.

### Arbitrary Drawing

As stated in the derivation, any $C^3$ function used for $x(t)$ and $y(t)$ will allow us to get an analytical solution. With that in mind we can draw arbitrary shapes by approximating them with quartic splines. Using MATLAB's *bwboundaries* function to detect edges of shapes, in combination with *spapi* which gives us an interpolating quartic spline given a set of points, we can draw the *TU Delft* logo[1].

[1]Different letter boundaries are connected by straight lines so the whole drawing can be one input function




(a) Reference and simulated trajectory.

(b) Open-loop motor voltages.

Figure 5: Open-loop drawing of the *TU Delft* logo with nominal model parameters.

Again, like with the ellipse, the logo, because of its sharp corners, is even more sensitive to inaccuracies in the model parameters. If we again perturb the parameters by ±5% we see that control of this form would not be useful anymore.

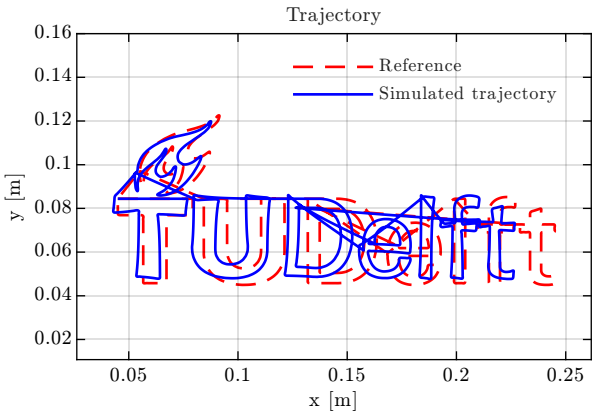



Figure 6: Open-loop drawing of the *TU Delft* logo with model parameters perturbed by ±5%.

### 1.3.1 Modelling the Nozzle

In the following section we model the nozzle of the 3D printer (Figure 7).

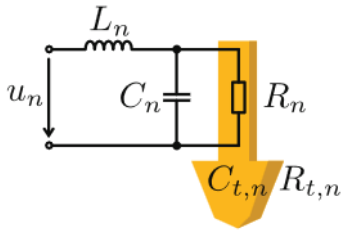



Figure 7: Schematic of the nozzle heating circuit.

add nicer drawing of the system

The system consists of an electrical and thermodynamical part. A circuit consisting of an inductor $L_n$, capacitor $C_n$ and a heating resistor $R_n$. We choose the current through the inductor and the voltage across the capacitor, $i_{L,n}$ and $u_c$ respectively, as first two states of our system. By solving the three Kirchhoff equations we get the following two first-order differential equations:

$$\begin{aligned} L_n \frac{di_{L,n}}{dt} &= u_n - u_c \\ C_n \frac{du_c}{dt} &= i_{L,n} - \frac{u_c}{R_n(T_r)} \end{aligned} \tag{1.27}$$

Next, we model the thermal part. Besides the thermal capacitance of the nozzle $C_{t,n}$, given in the task, we also model the thermal capacitance of the resistor $C_r$[2]. We choose two additional states, temperature of the resistor ($T_r$) and the temperature of the nozzle ($T_n$). Losses happening between the resistor and the nozzle are described by $R_{t,n}$, and the heat dissipation to the environment and to the filament is modelled with one variable $R_{\text{env}}$. The connection to the electrical part is the heat dissipated by the resistor $P_r = \frac{u_c^2}{R_n(T_r)}$. This gives us the following two equations which with (1.27) fully describe the system:

$$\begin{aligned} C_{t,n} \frac{dT_n}{dt} &= \frac{T_r - T_n}{R_{t,n}} - \frac{T_n - T_e}{R_{\text{env}}} \\ C_r \frac{dT_r}{dt} &= P_r - \frac{T_r - T_n}{R_{t,n}} \end{aligned} \tag{1.28}$$

with

$$R_n(T_r) = R_0 \left[1 + \alpha \left(T_r - T_0\right)\right] \tag{1.29}$$

where $R_0$ is the resistance at the reference temperature $T_0$ and $\alpha$ is the temperature coefficient of resistance.

Due to the square voltage term and the temperature dependence of the resistance the system is very non-linear, although the very different time constants of two systems (thermal states change much slower compared to the electrical) can be exploited. Alternatively, the model can reasonably be simplified by assuming a constant resistance. With that formulation we would get two linear systems connected by a single non-linear term.

[2] If we do not model the thermal capacitance of the resistor, we end up with a differential-algebraic equation (DAE).

### 1.3.2 Printing a circle of filament

**Moving-Boundary Heat Problem**

better notation, proper plots, explanation text

Cross-section $a \times a$, element length $\Delta L$; three faces to the environment ($T_e$), one face to the heating plate ($T_p$).

**Energy balance**

$$\partial_t u = -\nabla \cdot \phi + f, \qquad u = \rho c\, T, \qquad \phi = -\kappa \nabla T \qquad \Longrightarrow \qquad \partial_t T = \frac{\kappa}{\rho c}\, \nabla^2 T + \frac{f}{\rho c}$$

$$[u] = \mathrm{J/m^3}, \qquad [\phi] = \mathrm{W/m^2}, \qquad [f] = \mathrm{W/m^3}$$

**Heat exchange**

$$\Delta A_e = 3a\, \Delta L, \qquad \Delta A_p = a\, \Delta L, \qquad V = a^2\, \Delta L$$

$$\dot{Q} = K_e\, \Delta A_e\, (T_e - T) + K_p\, \Delta A_p\, (T_p - T) \qquad \left[\mathrm{W/(m^2 K)} \cdot \mathrm{m^2} \cdot \mathrm{K} = \mathrm{W}\right]$$

$$f = \frac{\dot{Q}}{V} = \frac{1}{a}\Big[\, 3K_e\, (T_e - T) + K_p\, (T_p - T)\,\Big]$$

**Result**

$$\boxed{\partial_t T = k\, \partial_{xx} T + \alpha\, T + \beta}$$

$$k = \frac{\kappa}{\rho c}, \qquad \alpha = -\frac{3K_e + K_p}{a\, \rho c}, \qquad \beta = \frac{3K_e T_e + K_p T_p}{a\, \rho c}$$

**2. Front Fixing**

John Crank, Free and Moving Boundary Problems

$$\xi = \frac{x}{s(t)}, \qquad 0 \le \xi \le 1, \qquad s(t) = s_0 + s_1 t, \qquad \dot{s} = s_1$$

**Derivatives**

$$\frac{\partial T}{\partial x} = \frac{1}{s}\, \frac{\partial T}{\partial \xi}, \qquad \frac{\partial^2 T}{\partial x^2} = \frac{1}{s^2}\, \frac{\partial^2 T}{\partial \xi^2}$$

$$\left(\frac{\partial T}{\partial t}\right)_x = \frac{\partial T}{\partial \xi}\left(\frac{\partial \xi}{\partial t}\right)_x + \left(\frac{\partial T}{\partial t}\right)_\xi = -\frac{x\, \dot{s}}{s^2}\, \frac{\partial T}{\partial \xi} + \left(\frac{\partial T}{\partial t}\right)_\xi = -\frac{\xi\, \dot{s}}{s}\, \frac{\partial T}{\partial \xi} + \left(\frac{\partial T}{\partial t}\right)_\xi$$

**Transformed equation**

$$\boxed{\left(\frac{\partial T}{\partial t}\right)_\xi = \frac{\xi\, \dot{s}}{s}\, \frac{\partial T}{\partial \xi} + \frac{k}{s^2}\, \frac{\partial^2 T}{\partial \xi^2} + \alpha\, T + \beta}$$

**3. Boundary and Initial Conditions**

$$T(1, t) = T_n \quad \text{(nozzle temperature at } x = s(t)\text{)}$$

$$\partial_x T(0, t) = 0 \quad \Longrightarrow \quad \frac{T_1^j - T_0^j}{\Delta \xi} = 0 \quad \Longrightarrow \quad T_0^j = T_1^j$$

$$T_i^0 = T_n, \qquad s(0) = s_0 \text{ (small)}$$

**4. Discretisation (implicit)**

$$\xi_i = i\,\Delta\xi, \quad \Delta\xi = \tfrac{1}{N}, \quad i = 0,\dots,N; \qquad t_j = j\,\Delta t; \qquad T_i^j \approx T(\xi_i, t_j)$$

$$\frac{\partial T}{\partial t} \approx \frac{T_i^j - T_i^{j-1}}{\Delta t}, \qquad \frac{\partial T}{\partial \xi} \approx \frac{T_{i+1}^j - T_i^j}{\Delta\xi}, \qquad \frac{\partial^2 T}{\partial \xi^2} \approx \frac{T_{i+1}^j - 2T_i^j + T_{i-1}^j}{\Delta\xi^2}$$

$$\gamma_i = \frac{\xi_i\,\dot{s}\,\Delta t}{s\,\Delta\xi}, \qquad g = \frac{k\,\Delta t}{s^2\,\Delta\xi^2}, \qquad s,\ \dot{s} \text{ evaluated at } t_j$$

$$\boxed{T_i^j\left[1 + \gamma_i + 2g - \alpha\,\Delta t\right] = g\,T_{i-1}^j + (\gamma_i + g)\,T_{i+1}^j + T_i^{j-1} + \beta\,\Delta t}$$

**Stability**

von Neumann probably

**5. Linear System at $t = t_j$**

$$d_i = 1 + \gamma_i + 2g - \alpha\,\Delta t, \qquad e_i = \gamma_i + g$$

$$\begin{bmatrix} 1 & -1 & & & & \\ -g & d_1 & -e_1 & & & \\ & -g & d_2 & -e_2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & -g & d_{N-1} & -e_{N-1} \\ & & & & 0 & 1 \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ T_2^j \\ \vdots \\ T_{N-1}^j \\ T_N^j \end{bmatrix} = \begin{bmatrix} 0 \\ T_1^{j-1} + \beta\,\Delta t \\ T_2^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-1}^{j-1} + \beta\,\Delta t \\ T_n \end{bmatrix}$$

**6. After the Boundary Stops (Closed Ring)**

$$\dot{s} = 0 \ \Rightarrow\ \gamma_i = 0, \qquad s = L, \qquad T_N^j \equiv T_0^j \text{ (first and last node coincide)}$$

$$d = 1 + 2g - \alpha\,\Delta t$$

$$\begin{bmatrix} d & -g & & & -g \\ -g & d & -g & & \\ & \ddots & \ddots & \ddots & \\ & & -g & d & -g \\ -g & & & -g & d \end{bmatrix} \begin{bmatrix} T_0^j \\ T_1^j \\ \vdots \\ T_{N-2}^j \\ T_{N-1}^j \end{bmatrix} = \begin{bmatrix} T_0^{j-1} + \beta\,\Delta t \\ T_1^{j-1} + \beta\,\Delta t \\ \vdots \\ T_{N-2}^{j-1} + \beta\,\Delta t \\ T_{N-1}^{j-1} + \beta\,\Delta t \end{bmatrix}$$

First step after connection:

$$T_0^{j-1} \leftarrow \tfrac{1}{2}\left(T_0^{j-1} + T_N^{j-1}\right)$$

Choose and display the chosen model parameters

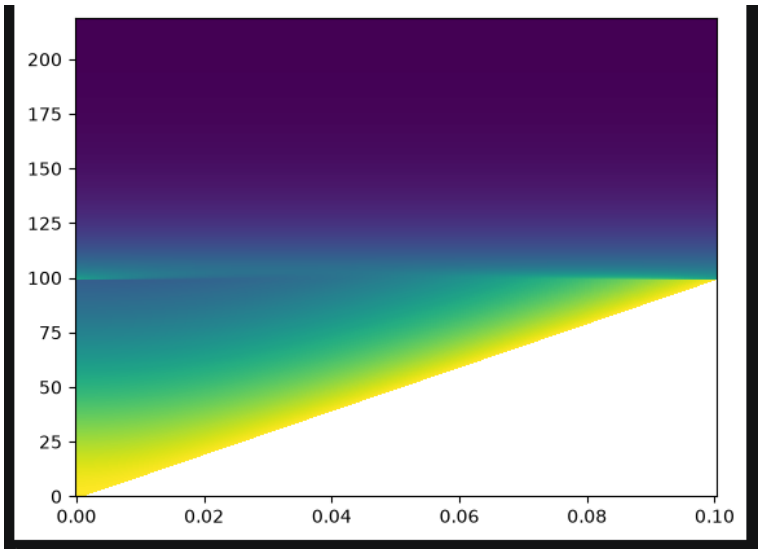



Figure 8: INITIAL PLOT, CHANGE.

# 2 Finite State Machine

## 2.1 Filament Replacement

The filament replacement is modelled with two machines: a printing machine and a filament replacement machine.

### 2.1.1 Formal Machine Definition

The FSM is defined as the Mealy machine

$$\mathcal{M} = (Q, \Sigma, \Lambda, \delta, \lambda, q_0). \tag{2.1}$$

**States.** The state $q = (p, f, r)$ is the printing state, the filament replacement state and the retries left,

$$\begin{aligned} Q &= P \times \mathcal{F} \times R, \\ P &= \{\texttt{IDLE}, \texttt{IN_PROG}, \texttt{PAUSED}\}, \\ \mathcal{F} &= \{\texttt{IDLE}, \texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}, \\ R &= \{0, 1, 2, 3\}, \end{aligned} \tag{2.2}$$

where `MOVE` moves to the wipe location, `WIPE` wipes the excess filament, `COOL` switches the nozzle heat off, `LOAD` loads the new filament and `HEAT` switches the nozzle heat on. The head location is saved on entering `MOVE` and the print is resumed on leaving `HEAT`. The initial state is $q_0 = (\texttt{IDLE}, \texttt{IDLE}, 3)$.

**Inputs.** An input is a pair $\sigma = (e, y)$. Here $e$ is the external event ($C$, $F$ or nothing) and $y$ is the feedback of the current step: $\text{ok}_k$ or $\text{fail}_k$ for each step $k \in K = \{\texttt{MOVE}, \texttt{WIPE}, \texttt{COOL}, \texttt{LOAD}, \texttt{HEAT}\}$, or nothing. This gives $3 \cdot 11 = 33$ inputs:

$$\begin{aligned} E &= \{C,\ F,\ \varepsilon\}, \\ Y &= \{\varepsilon,\ \text{ok}_{\texttt{MOVE}},\ \text{fail}_{\texttt{MOVE}},\ \text{ok}_{\texttt{WIPE}},\ \text{fail}_{\texttt{WIPE}},\ \ldots,\ \text{ok}_{\texttt{HEAT}},\ \text{fail}_{\texttt{HEAT}}\}, \\ \Sigma &= E \times Y. \end{aligned} \tag{2.3}$$

**Outputs.** The outputs $\Lambda$ are messages to the user, for example "filament change in progress", "$k$ successful", "$k$ failure, trying again", "filament change process failed, initiate filament change again" and "filament change already in progress".

**Transition and output functions.** The transition function $\delta : Q \times \Sigma \to Q$ and output function $\lambda : Q \times \Sigma \to \Lambda$ are given by Table 2 and Figures 9 and 10.

**Implementation.** The machine is a transition table (Table 2) in `fsm.ipynb`. The rows matching the current state and input are checked and the corresponding outputs are determined. `X` means `DON'T CARE`.

Table 2: Snapshot of the state transition table (state names abbreviated, `X` is a `DON'T CARE`, empty cells mean no signal).

| $p$ | $f$ | $e$ | $y$ | $r$ | $p'$ | $f'$ | message | $r'$ |
|---|---|---|---|---|---|---|---|---|
| IDLE | IDLE | C | | X | IDLE | MOVE | filament change in progress | 3 |
| IN_PROG | IDLE | C | | X | PAUSED | MOVE | filament change in progress | 3 |
| X | MOVE | | ok_MOVE | X | X | WIPE | moving to wiping location successful | 3 |
| X | MOVE | | fail_MOVE | 3 | X | MOVE | failure, trying again | 2 |
| X | MOVE | | fail_MOVE | 0 | X | IDLE | process failed, initiate again | 3 |
| PAUSED | HEAT | | ok_HEAT | X | IN_PROG | IDLE | nozzle heating successful | 3 |
| X | WIPE | C | X | X | X | WIPE | filament change already in progress | X |

The model was checked against a set of different test cases (`test_cases.csv`).

### 2.1.2 Transition

The two sub-machines are shown in Figures 9 and 10. Transitions are labelled *input* [*guard*] / *action*, where $r$ is the number of retries left. A $C$ or $F$ event starts a replacement only when the filament machine is idle; otherwise it is ignored. If it arrives together with a feedback signal, the event takes precedence.

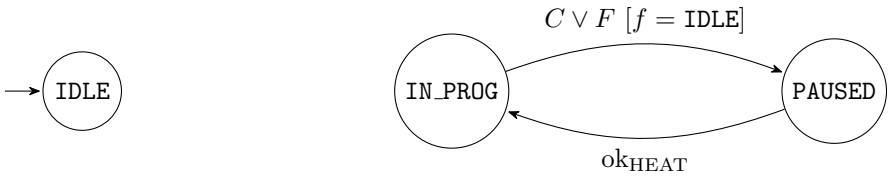



Figure 9: Printing sub-machine with state $p \in P$. The transition from `IDLE` to `IN_PROG` (print started) belongs to the print job handling and is not modelled here.

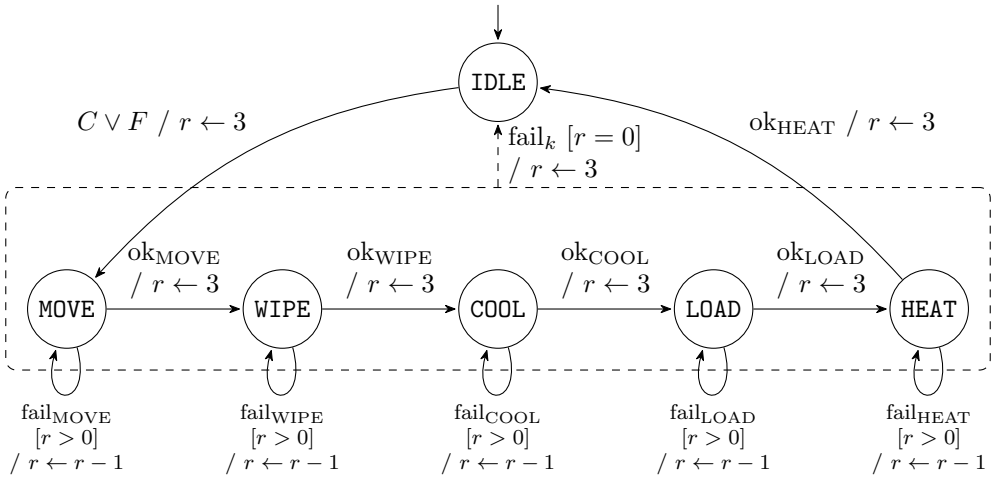



Figure 10: Filament replacement sub-machine with state $f \in \mathcal{F}$. For readability the five $\mathrm{fail}_k\,[r = 0]$ transitions from the steps $k$ back to IDLE are drawn as a single dashed arrow from the box around the steps. The two sub-machines are synchronised: the $C \vee F$ event and the $\mathrm{ok}_{\mathrm{HEAT}}$ event act on both machines in the same step.

## 2.2 Parked Detection

The head is parked at the origin when both motions have hit their end of line, i.e. a motion was commanded but the motor did not rotate. This is detected by the DFA

$$\mathcal{A} = (Q_A, \Sigma_A, \delta_A, q_{A,0}, F_A) \tag{2.4}$$

with

$$\begin{aligned} Q_A &= \{q_0, q_1, q_2, q_3\}, \\ \Sigma_A &= M \times m = \{(H,+), (H,0), (V,+), (V,0)\}, \\ q_{A,0} &= q_0, \qquad F_A = \{q_3\}, \end{aligned} \tag{2.5}$$

where the states have the meaning

- $q_0$: no end of line detected,
- $q_1$: horizontal end of line (left side) detected,
- $q_2$: vertical end of line (bottom side) detected,
- $q_3$: both detected, the head is parked.

The transition function $\delta_A : Q_A \times \Sigma_A \to Q_A$ is given in Table 3 and drawn in Figure 11.

Table 3: Transition function $\delta_A$ of the parked detector.

| $q$ | $(H,+)$ | $(H,0)$ | $(V,+)$ | $(V,0)$ |
|---|---|---|---|---|
| $\to q_0$ | $q_0$ | $q_1$ | $q_0$ | $q_2$ |
| $q_1$ | $q_0$ | $q_1$ | $q_1$ | $q_3$ |
| $q_2$ | $q_2$ | $q_3$ | $q_0$ | $q_2$ |
| $* q_3$ | $q_2$ | $q_3$ | $q_1$ | $q_3$ |

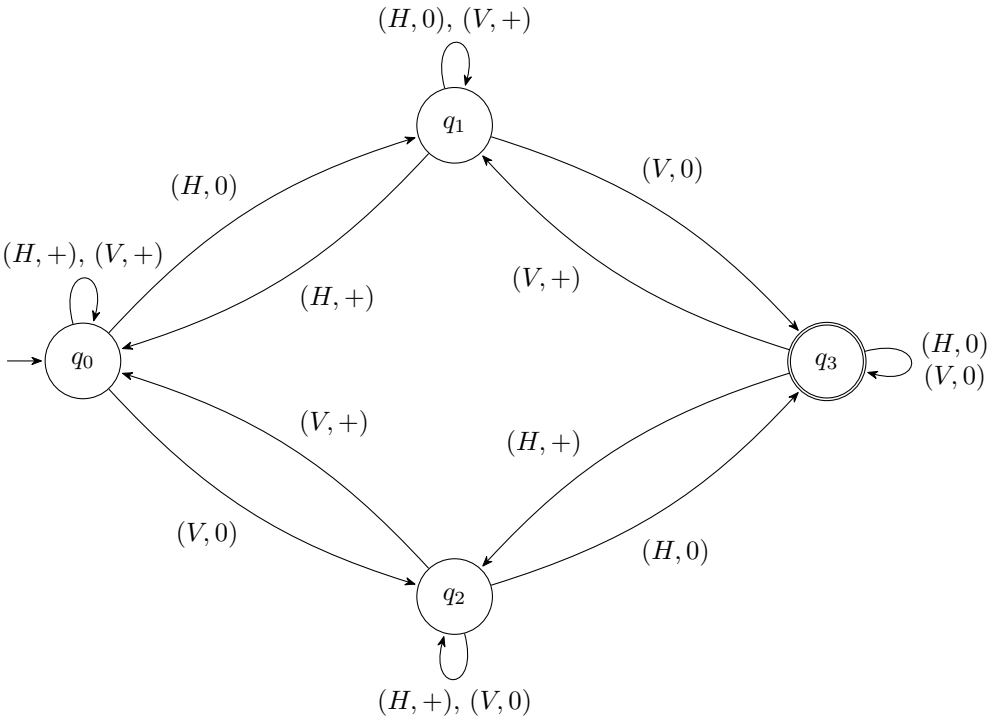



Figure 11: Transition of the parked detector $\mathcal{A}$. The accepted language is the set of sequences $\sigma$ that end in $q_3$.

## 2.3 Detecting the Other Corners

The automaton is extended to the motions $H_x$, $H_y$ and $V$ and to the angular velocities $+$, $-$ and $0$,

$$\begin{aligned} \mathcal{A}' &= (Q_A, \Sigma'_A, \delta'_A, q_0, F_A), \\ \Sigma'_A &= \{H_x, H_y, V\} \times \{+, -, 0\}, \end{aligned} \tag{2.6}$$

with $Q_A$, $q_0$ and $F_A$ as before. The transition function $\delta'_A : Q_A \times \Sigma'_A \to Q_A$ is given in Table 4 and drawn in Figure 12.

Table 4: Transition function $\delta'_A$ of the corner detector.

| $q$ | $(H_x, \pm)$ | $(H_x, 0)$ | $(H_y, \pm)$ | $(H_y, 0)$ | $(V, \cdot)$ |
|---|---|---|---|---|---|
| $\to q_0$ | $q_0$ | $q_1$ | $q_0$ | $q_2$ | $q_0$ |
| $q_1$ | $q_0$ | $q_1$ | $q_1$ | $q_3$ | $q_1$ |
| $q_2$ | $q_2$ | $q_3$ | $q_0$ | $q_2$ | $q_2$ |
| $* q_3$ | $q_2$ | $q_3$ | $q_1$ | $q_3$ | $q_3$ |

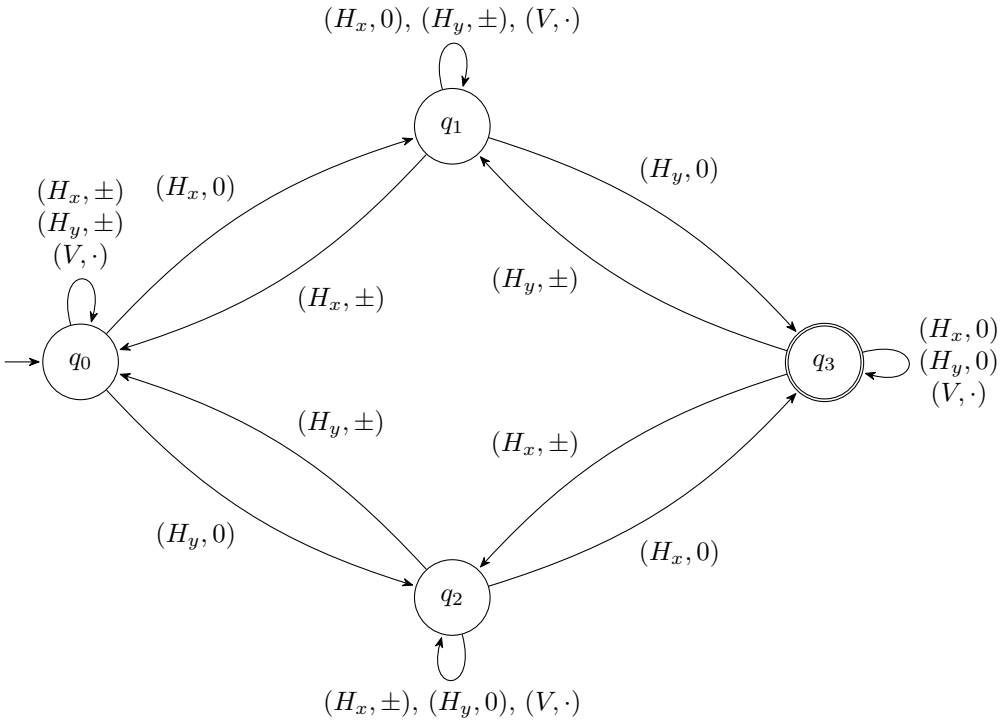



Figure 12: Transition of the corner detector $\mathcal{A}'$.

# 3 Hybrid System

## 3.1 Part 1: Formal Definition

We define the hybrid state machine as

$$H = (Q, X, U, Y, f, h, \mathit{Init}, E, G, R).$$

It combines three smaller machines running in parallel: the nozzle machine $\mathcal{N}$, the printer machine $\mathcal{P}$ and the filament machine $\mathcal{M}$.

In mode $q$ the continuous state flows as $\dot{x} = f(q, x, u)$. An edge is taken as soon as its guard is true, and $z^+$ denotes the value of $z$ right after the transition. There are no separate invariants.

**$Q$: set of modes.** One mode from each machine, $Q = Q_{\mathcal{N}} \times Q_{\mathcal{P}} \times Q_{\mathcal{M}}$:

$$\begin{aligned} Q_{\mathcal{N}} &= \{\text{COOL}, \text{HEATING_UP}, \text{HEATED}, \text{COOLING}\}, \\ Q_{\mathcal{P}} &= \{\text{IDLE}, \text{WAIT_HOT}, \text{TRAVEL_TO_START}, \text{PRINT_PATH}, \\ &\qquad \text{PAUSED}, \text{COMPLETED}\}, \\ Q_{\mathcal{M}} &= \{\text{IDLE}, \text{MOVING_TO_WIPE}, \text{WIPING}, \text{NOZZLE_COOLING}, \\ &\qquad \text{LOADING_FILAMENT}, \text{NOZZLE_HEATING}, \text{READY_TO_RESUME}\}. \end{aligned}$$

WAIT_HOT, TRAVEL_TO_START and PRINT_PATH together form the mode IN_PROGRESS of $\mathcal{P}$.

**$X$: continuous state.** The printer state $x_p = (x, y, \dot{x}, \dot{y}, T_n, T_r)$ (head position and velocity, nozzle and heater temperatures) plus the variables of Table 5. We use the reduced plant, where the fast electrical dynamics are instantaneous ($L_m \approx 0$, $u_c = u_n$).

Table 5: Variables in $X$ besides the printer state $x_p$.

| Symbol | Meaning |
|---|---|
| $L$ | Length of filament used so far |
| $\tau$ | Time spent in the current mode; a timer with $\dot{\tau} = 1$ that is reset to 0 on every edge |
| $k$ | Index of the current point in the path, which is an array of coordinates that the printer follows; $k \in \{1, \dots, n\}$ |
| $(x_s, y_s)$ | Coordinates of the head saved when the printer is paused |
| *req*, *heated* | Request to the nozzle to start heating; indicator that the nozzle is heated |
| *paused* | Indicator that the printer is in the paused state |
| *change_active*, *change_done* | Indicator that a filament change is in progress; indicator that a filament change has finished |

**$U = U_D \times U_C$: inputs.** $U_D = \{C, START\}$ are the user events (colour change, start). The continuous inputs $U_C$ are listed in Table 6. They are fixed by the mode, as given in Section 3.2. The motors are driven by $\mathcal{M}$ when it is not IDLE and by $\mathcal{P}$ otherwise.

Table 6: Continuous inputs $U_C$.

| Symbol | Code variable | Meaning |
|---|---|---|
| $(u_{mL}, u_{mR}, u_n)$ | `mode_motor`, `mode_nozzle` | Left and right motor voltages and heater voltage; each is either off or on, depending on the mode |
| Extruder switch | `is_extruding` | Whether the extruder is on |
| $r = (x_c, y_c)$ | `command_x`, `command_y` | Commanded head position that the motors drive towards |

**$Y$: outputs.** The outputs are the head position $(x, y)$, the nozzle temperature $T_n$, the length of filament used $L$, and the current mode of each machine $(q_{\mathcal{N}}, q_{\mathcal{P}}, q_{\mathcal{M}})$. Here the mode of a machine is the step it is currently in, for example HEATED for $\mathcal{N}$, PAUSED for $\mathcal{P}$ or WIPING for $\mathcal{M}$.

**$f$: flow map.** The reduced CoreXY plant (`sys_reduced`): a damped head mass driven by the two coupled motors, and a two-node nozzle thermal model heated with power $u_n^2/R_n(T_r)$. The other variables follow Table 5.

**$h$: output map.** $h$ computes the outputs from the current mode $q = (q_{\mathcal{N}}, q_{\mathcal{P}}, q_{\mathcal{M}})$ and continuous state $x$, so $y_{out} = h(q, x)$. It does not change the values; it only selects which quantities are observed:

$$y_{out} = h(q, x) = \big( x,\ y,\ T_n,\ L,\ q_{\mathcal{N}},\ q_{\mathcal{P}},\ q_{\mathcal{M}} \big).$$

The first four entries are the head position $(x, y)$, $T_n$ and $L$, read directly from $x_p$ and $L$. The last three come from $q$. The remaining variables ($\dot{x}$, $\dot{y}$, $T_r$, $\tau$, $k$, the saved coordinates and the flags) are internal and not part of the output.

*Init*: **initial state.** $\mathit{Init} = \{(q_0, x_0)\}$ contains a single element:

$$\begin{aligned} q_0 &= (\text{COOL}, \text{IDLE}, \text{IDLE}), \\ x_p(0) &= (x, y, \dot{x}, \dot{y}, T_n, T_r) = (0, 0, 0, 0, T_e, T_e), \\ (L, \tau, k) &= (0, 0, 1), \\ (\mathit{req}, \mathit{heated}, \mathit{paused}, \mathit{change_active}, \mathit{change_done}) &= (\text{false}, \text{false}, \text{false}, \text{false}, \text{false}), \end{aligned}$$

where $T_e$ is the ambient temperature.

**Shorthand for the guards.** The conditions in Table 7 are used to keep the guards short.

Table 7: Shorthand used in the guards.

| Symbol | Meaning |
|---|---|
| $\mathit{arr}$ | Head has arrived at its target: $\lvert x - x_c\rvert < \varepsilon$ and $\lvert y - y_c\rvert < \varepsilon$, with $\varepsilon = 0.5\,\text{mm}$ |
| $F$ | Spool is empty: $L \geq L_{spool}$ |
| $T_{hi}$ | Temperature at which the nozzle counts as hot, $T_{hi} = 350\,\text{K}$ |
| $\mathit{arr}_{>0.5}$ | $\mathit{arr}$ has held for more than 0.5 s |
| $(T_n \geq T_{hi})_{>1}$ | $T_n \geq T_{hi}$ has held for more than 1 s |
| $C^{\uparrow}$ | Button $C$ has just been pressed (rising edge, $0 \to 1$) |

$E$: **edges.** $E \subseteq Q \times Q$ is the set of allowed transitions $e = (q, q')$. It is the union of the edges of the three machines, which are listed in Section 3.2:

$$\begin{aligned} E &= E_{\mathcal{N}} \cup E_{\mathcal{P}} \cup E_{\mathcal{M}}, \\ e &= (q, q'), \quad \text{e.g. } e = (\text{COOL}, \text{HEATING_UP}) \in E_{\mathcal{N}}. \end{aligned}$$

$G$: **guards.** $G(e)$ is the condition on $x$ and $u$ under which the edge $e$ is taken:

$$\begin{aligned} G(\text{COOL}, \text{HEATING_UP}) &= \mathit{req}, \\ G(\text{HEATING_UP}, \text{HEATED}) &= (T_n \geq T_{hi})_{>1}. \end{aligned}$$

The edges IN_PROGRESS $\to$ PAUSED in $\mathcal{P}$ and IDLE $\to$ MOVING_TO_WIPE in $\mathcal{M}$ share the guard $C^{\uparrow} \vee F$, so they fire together.

$R$: **resets.** $R(e)$ gives the values $z^+$ of the variables right after $e$ is taken:

$$\begin{aligned} R(\text{IDLE}, \text{WAIT_HOT}) &: \ \mathit{req}^+ = \text{true}, \\ R(\text{PRINT_PATH}, \text{PRINT_PATH}) &: \ k^+ = k + 1. \end{aligned}$$

Every edge also resets $\tau^+ = 0$ (not repeated in the tables), and any variable not listed keeps its value, $z^+ = z$.

## 3.2 Part 2: The Machines

Each machine is shown as a flowchart that gives its modes, the inputs $u$ used in each mode, and the edges with their guard $G$ and reset $R$, followed by a short explanation.

### 3.2.1 Nozzle Machine $\mathcal{N}$

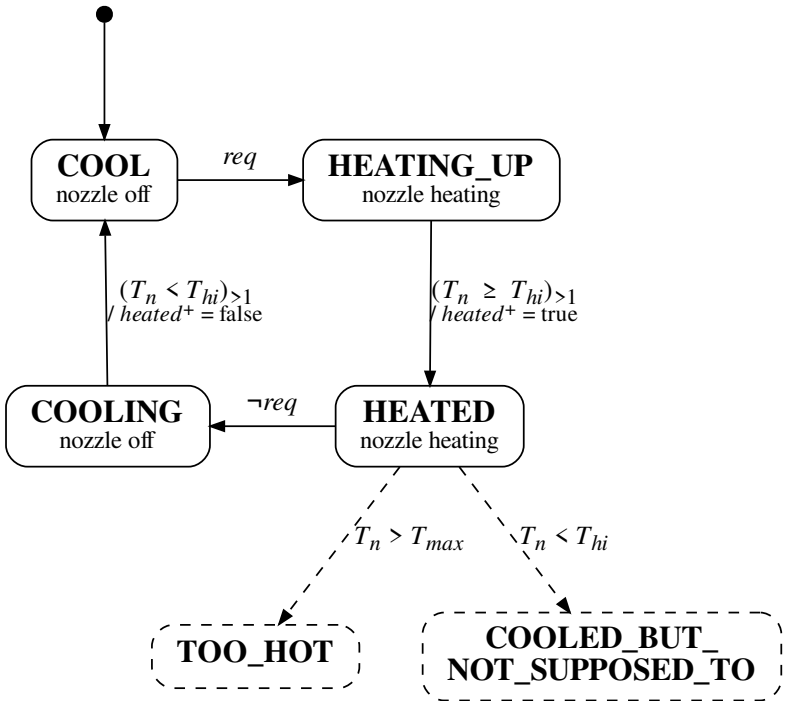



Figure 13: Nozzle machine, in the notation of this report. The dashed modes are failure modes of the implementation, which are not part of $H$. Edge labels show the guard $G$, and below it the reset $R$ after the slash. Inputs $u$ are given in each mode.

- The nozzle starts in COOL with the heater off. When heating is requested (*req*), it moves to HEATING_UP and heats.
- After $T_n \geq T_{hi}$ has held for more than 1 s it is HEATED and *heated* becomes true. The machine keeps heating and does not regulate $T_n$ further.
- When *req* is withdrawn the heater is switched off (COOLING), and after $T_n < T_{hi}$ for more than 1 s it returns to COOL with *heated* false.

### 3.2.2 Printer Machine $\mathcal{P}$

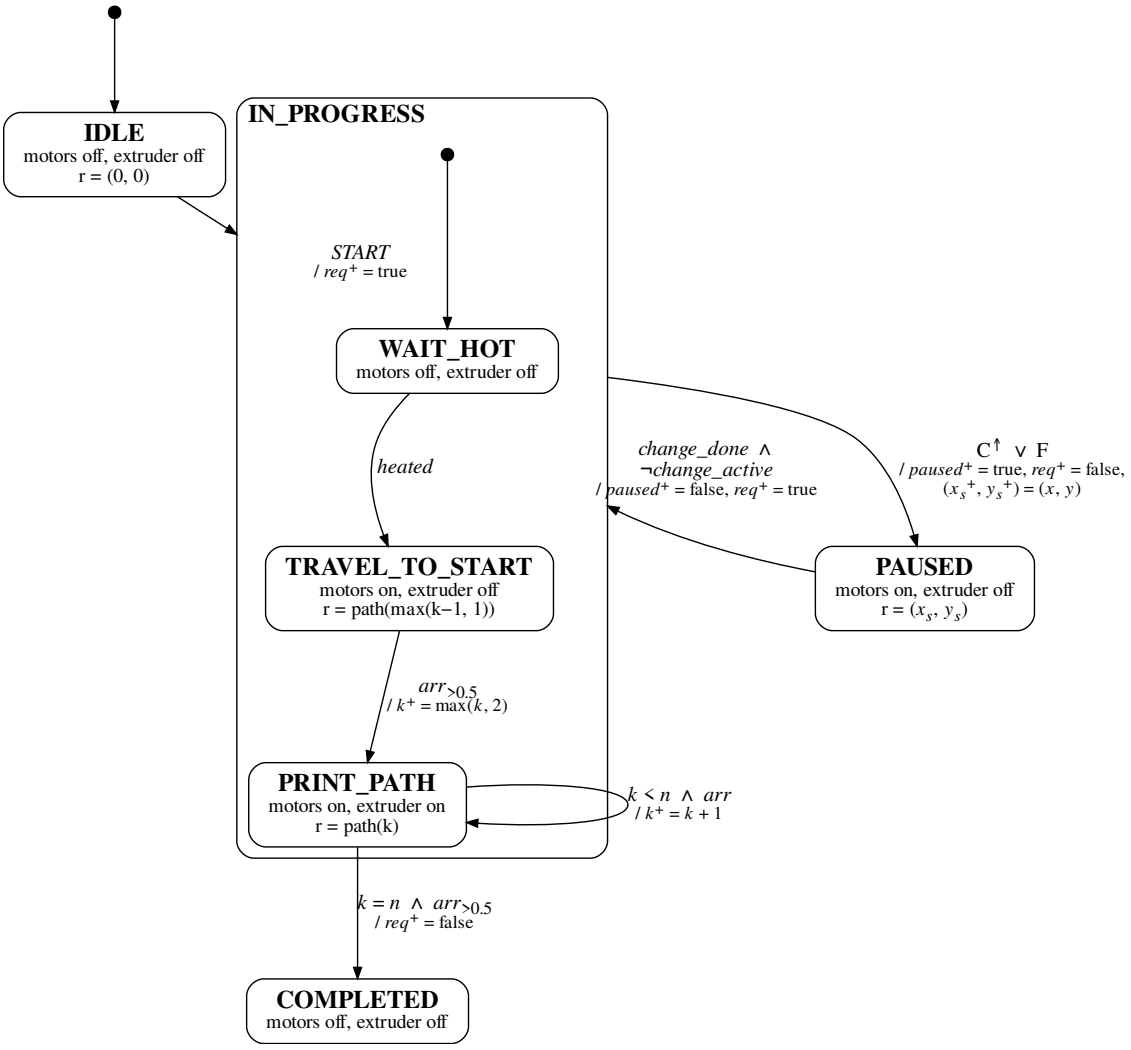



Figure 14: Printer machine, in the notation of this report. Edge labels show the guard $G$, and below it the reset $R$ after the slash. Inputs $u$ are given in each mode.

- *START* requests heating and the printer waits in WAIT_HOT until the nozzle is *heated*.
- The head then travels to the start of the path and prints it point by point, increasing $k$ each time it arrives. After the last point the printer is COMPLETED.
- On $C^{\uparrow}$ or $F$ the printer pauses from any sub-mode of IN_PROGRESS and saves the head position $(x_s, y_s)$. When the change is done it re-enters WAIT_HOT, so it waits for the nozzle, travels back to path$(k-1)$ and continues from point $k$.

### 3.2.3 Filament Machine $\mathcal{M}$

The wipe location is $(x_w, y_w)$, the wipe time is $t_w$ and the load time is $t_l$.

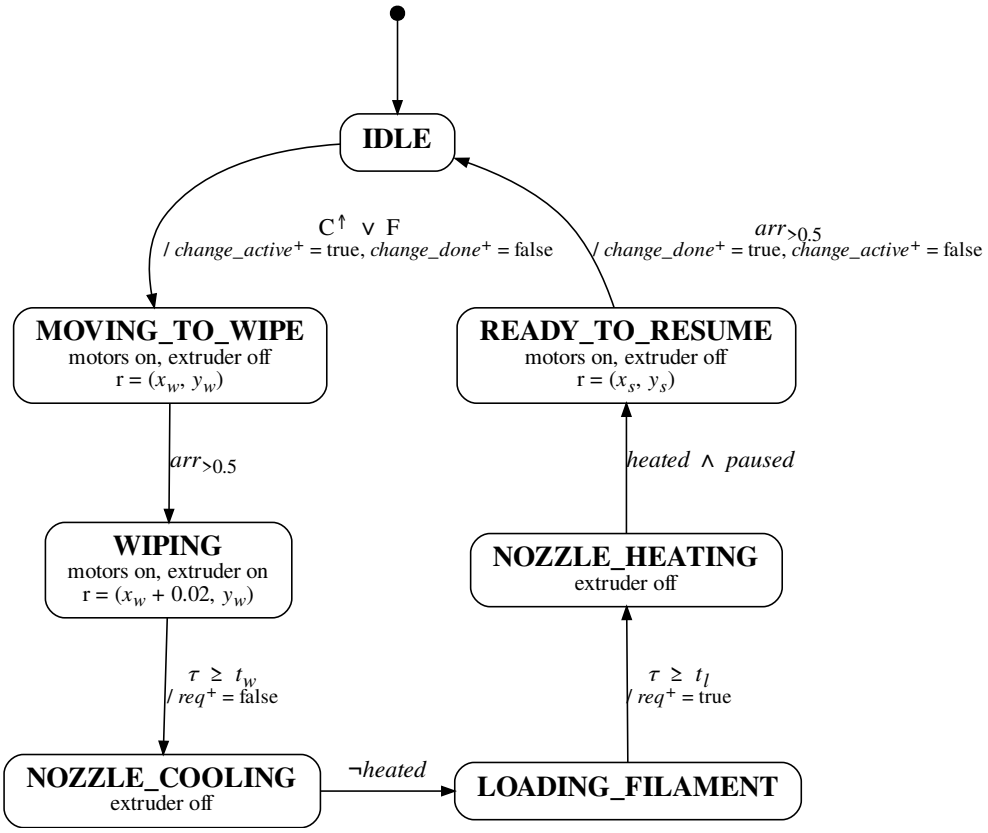



Figure 15: Filament machine, in the notation of this report. Edge labels show the guard $G$, and below it the reset $R$ after the slash. Inputs $u$ are given in each mode.

- On $C^{\uparrow}$ or $F$ the change starts: the head moves to the wipe location $(x_w, y_w)$ and wipes with the extruder on for $t_w = 3\,\mathrm{s}$.
- Heating is then switched off (*req* false). Once the nozzle has cooled ($\neg\, heated$), new filament is loaded for $t_l = 5\,\mathrm{s}$.
- Heating is requested again. When the nozzle is *heated* and the printer is paused, the head returns to $(x_s, y_s)$ and *change_done* becomes true, which releases the printer.

**Note.** The nozzle flowchart shows two failure modes of the implementation (dashed). The implementation also has a *FAILED* mode and a manual pause and resume input. The definitions above ignore these failure modes and all retries, and cover only the successful filament change.

### 3.3 Part 3: Drawing a Square

**Path Tracking** The square is drawn from a list of points $\mathrm{path}(k) = (x_k, y_k)$, $k = 1, \ldots, n$, with $n = 25$.

- The path runs counter-clockwise through the corners $(0.05, 0.05)$, $(0.25, 0.05)$, $(0.25, 0.25)$, $(0.05, 0.25)$ (in metres) and back to the start.

- Each side has five intermediate points, so consecutive points are 33.3 mm apart.
- In PRINT_PATH the head is commanded to path($k$) with the extruder on.
- Once *arr* holds, the index is increased to $k + 1$ and the head moves to the next point.
- The square is therefore drawn point by point, until $k = n$.

**Cases.** The same square is drawn twice with a filament change in the middle of the print:

1. $C$ is triggered half-way through the print;
2. $F$ happens: the spool is empty half-way through the print.

In both cases the head is moved to the wipe location $(0, 0.5)$ m, and the part printed before the change is shown in red and the part after it in green.

### 3.3.1 Case 1: $C$ Triggered

$C$ is pressed at $t \approx 69.6$ s, half-way through the print (after two sides of the square). The printer pauses, the head wipes at $(0, 0.5)$ m, the filament is loaded and the nozzle reheats, and printing resumes at $t \approx 96.0$ s (red to green). The square is complete at $t \approx 163$ s.

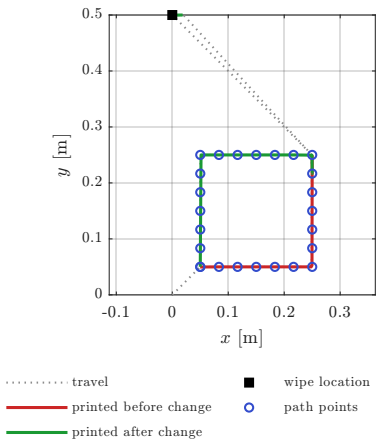



Figure 16: Case 1, head path: printed before the change (red), after (green), and travel moves.

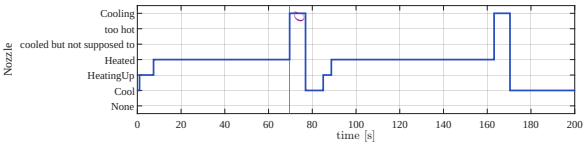



Figure 18: Case 1, nozzle machine mode.

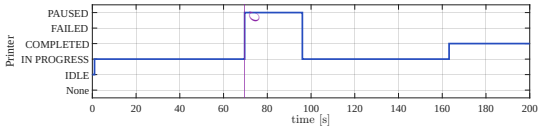



Figure 19: Case 1, printer machine mode.

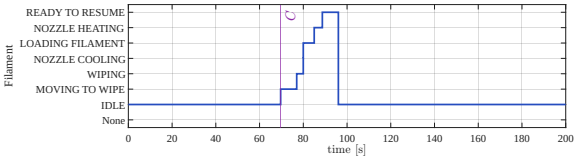



Figure 20: Case 1, filament machine mode.

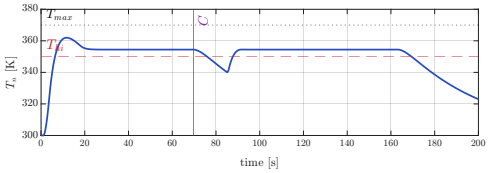



Figure 17: Case 1, nozzle temperature $T_n$ with the thresholds $T_{hi}$ and $T_{max}$.

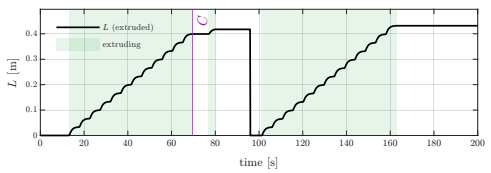



Figure 21: Case 1, extruded filament length $L$ against $L_{spool}$.

### 3.3.2 Case 2: $F$ Triggered

$C$ is never pressed. The spool holds half of the filament needed for the square, so $F$ occurs at $t \approx 69.6\,\text{s}$ and the same change sequence runs, resuming at $t \approx 96.0\,\text{s}$. The replacement spool has the same length, so $F$ occurs a second time at $t \approx 158\,\text{s}$ (resuming at 185.6 s) and the square is complete at $t \approx 191\,\text{s}$.

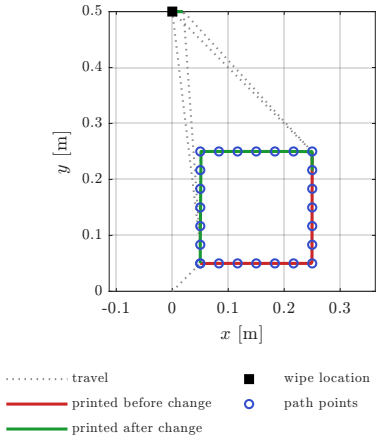



Figure 22: Case 2, head path: printed before the change (red), after (green), and travel moves.

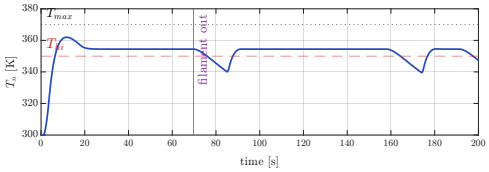



Figure 23: Case 2, nozzle temperature $T_n$ with the thresholds $T_{hi}$ and $T_{max}$.

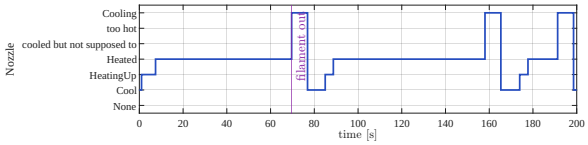



Figure 24: Case 2, nozzle machine mode.

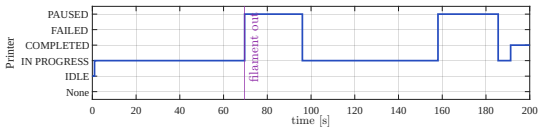



Figure 25: Case 2, printer machine mode.

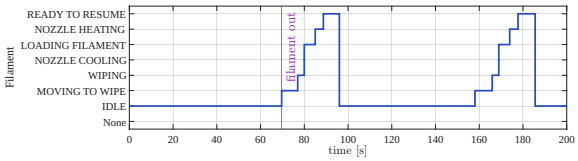



Figure 26: Case 2, filament machine mode.

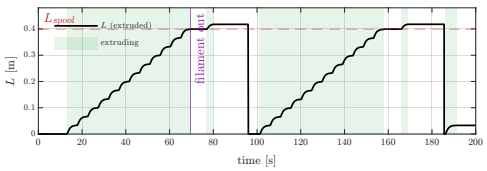



Figure 27: Case 2, extruded filament length $L$ against $L_{spool}$.

# References